

The matrix of the linear operator $f : \mathbb{R}^{10} \rightarrow \mathbb{R}^{10}$ with regard to the basis $B = (\vec{b}_1, \dots, \vec{b}_{10})$ is in a Jordan normal form with three blocks of eigenvalue 0 with sizes of 5, 3, 2. For all $i, j \in \mathbb{N}$ find using the basis B some basis of the subspace

$$(\text{Ker } f^i) \cap (\text{Im } f^j)$$

Rewriting the first sentence we get

$$[f]_B^B = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & & & & & \\ 0 & 0 & 1 & 0 & 0 & & & & & \\ 0 & 0 & 0 & 1 & 0 & & 0 & & & \\ 0 & 0 & 0 & 0 & 1 & & & & & \\ 0 & 0 & 0 & 0 & 0 & & & & & \\ & & & & & 0 & 1 & 0 & & \\ & & & & & 0 & 0 & 1 & 0 & \\ & & & & & 0 & 0 & 0 & & \\ & & & & & & & & 0 & 1 \\ & & & & & & & & 0 & 0 \end{pmatrix}$$

Because f has an eigenvalue of 0 with an algebraic multiplicity of 10 we can define

$$g = f - \lambda \text{id} = f - 0 \text{id} = f$$

Then if $f(\vec{v}) = \lambda \cdot \vec{v}$ then $g(\vec{v}) = \vec{0}$. Since there are three jordan blocks, the geometric multiplicity is 3. We will therefore get three jordan chains

$$\begin{aligned} \vec{b}_5 &\xrightarrow{g} \vec{b}_4 \xrightarrow{g} \vec{b}_3 \xrightarrow{g} \vec{b}_2 \xrightarrow{g} \vec{b}_1 \xrightarrow{g} \vec{0} \\ \vec{b}_8 &\xrightarrow{g} \vec{b}_7 \xrightarrow{g} \vec{b}_6 \xrightarrow{g} \vec{0} \\ \vec{b}_{10} &\xrightarrow{g} \vec{b}_9 \xrightarrow{g} \vec{0} \end{aligned}$$

Since $f(t\vec{v}) = t \cdot f(\vec{v})$ we get and any $\vec{v} \in \mathbb{R}^{10}$ can be expressed as a linear combination of elements of B , the following chart describes the behaviour of all vectors in $\mathbb{R}^{10}$

$$\begin{aligned} \text{Span}\{\vec{b}_5\} &\xrightarrow{f} \text{Span}\{\vec{b}_4\} \xrightarrow{f} \text{Span}\{\vec{b}_3\} \xrightarrow{f} \text{Span}\{\vec{b}_2\} \xrightarrow{f} \text{Span}\{\vec{b}_1\} \xrightarrow{f} \vec{0} \\ \text{Span}\{\vec{b}_8\} &\xrightarrow{f} \text{Span}\{\vec{b}_7\} \xrightarrow{f} \text{Span}\{\vec{b}_6\} \xrightarrow{f} \vec{0} \\ \text{Span}\{\vec{b}_{10}\} &\xrightarrow{f} \text{Span}\{\vec{b}_9\} \xrightarrow{f} \vec{0} \end{aligned}$$

Since both $\text{Span}\{\vec{b}_1\}$, $\text{Span}\{\vec{b}_6\}$ and $\text{Span}\{\vec{b}_9\}$ all get mapped to $\vec{0}$ any linear combination of $\vec{b}_1, \vec{b}_6$ and $\vec{b}_9$ will also get mapped to $\vec{0}$, hence $\text{Ker } f = \text{Span}\{\vec{b}_1, \vec{b}_6, \vec{b}_9\}$. Importantly, if the linear combination contains any other $\vec{b}_i$ it will not get mapped to $\vec{0}$ but rather some other linear combination of $\vec{b}_j$ with the same coefficients. (this we get from linearity of f) Since they are linearly independent, it will

always be non zero. Through similar reasoning we get

$$\begin{aligned}
\text{Ker } f &= \text{Span}\{\vec{b}_1, \vec{b}_6, \vec{b}_9\} \\
\text{Ker } f^2 &= \text{Span}\{\vec{b}_1, \vec{b}_6, \vec{b}_9, \vec{b}_2, \vec{b}_7, \vec{b}_{10}\} \\
\text{Ker } f^3 &= \text{Span}\{\vec{b}_1, \vec{b}_6, \vec{b}_9, \vec{b}_2, \vec{b}_7, \vec{b}_{10}, \vec{b}_3, \vec{b}_8\} \\
\text{Ker } f^4 &= \text{Span}\{\vec{b}_1, \vec{b}_6, \vec{b}_9, \vec{b}_2, \vec{b}_7, \vec{b}_{10}, \vec{b}_3, \vec{b}_8, \vec{b}_4\} \\
\text{Ker } f^5 &= \text{Span}\{\vec{b}_1, \vec{b}_6, \vec{b}_9, \vec{b}_2, \vec{b}_7, \vec{b}_{10}, \vec{b}_3, \vec{b}_8, \vec{b}_4, \vec{b}_5\} = \text{Span } B
\end{aligned}$$

Using the same ideas on $\text{Im } f$ we get

$$\begin{aligned}
\text{Im } f &= \text{Span}\{\vec{b}_4, \vec{b}_7, \vec{b}_9, \vec{b}_3, \vec{b}_6, \vec{b}_2, \vec{b}_1\} \\
\text{Im } f^2 &= \text{Span}\{\vec{b}_3, \vec{b}_6, \vec{b}_2, \vec{b}_1\} \\
\text{Im } f^3 &= \text{Span}\{\vec{b}_2, \vec{b}_1\} \\
\text{Im } f^4 &= \text{Span}\{\vec{b}_1\} \\
\text{Im } f^5 &= \{\vec{0}\}
\end{aligned}$$

From which we get (the windows contain the basis of the intersections)

$\cap$	$\text{Ker } f$	$\text{Ker } f^2$	$\text{Ker } f^3$	$\text{Ker } f^4$	$\text{Ker } f^n, n \geq 5$
$\text{Im } f$	b_1, b_6, b_9	b_1, b_6, b_9, b_2, b_7	$b_1, b_6, b_9, b_2, b_7, b_3$	$b_1, b_6, b_9, b_2, b_7, b_3, b_4$	$\text{Im } f$
$\text{Im } f^2$	b_1, b_6	b_1, b_6, b_2	b_1, b_6, b_2, b_3	b_1, b_6, b_2, b_3	$\text{Im } f^2$
$\text{Im } f^3$	b_1	b_1, b_2	b_1, b_2	b_1, b_2	$\text{Im } f^3$
$\text{Im } f^4$	b_1	b_1	b_1	b_1	$\text{Im } f^4$
$\text{Im } f^n, n \geq 5$	$\{\vec{0}\}$	$\{\vec{0}\}$	$\{\vec{0}\}$	$\{\vec{0}\}$	$\{\vec{0}\}$

Q. E. D.